

SE LAB 2

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Section: B

Title: Agile Backlog Creation & Sprint Simulation in Jira

Scenario (from Lab 1): Automated Rubric Assignment Evaluator

Jira project: Rubric Assignment Evaluator - Lab 2 (ARAE), company-managed Scrum

1. Epics and User Stories

Epic 1: Project Submission (ARAE-1)

Enable students to upload their project submissions as ZIP files and receive confirmation so the work can enter automated evaluation. (Lab 1: FR-002, UC-01)

Key	User Story	As a	I want	So that	Priority	Story Points
ARAE-6	Story 1.1: Upload Project ZIP	student	to upload my project submission as a ZIP file	it can be evaluated automatically	High	5
ARAE-7	Story 1.2: Submission Confirmation	student	to receive a confirmation after my upload completes	I know my submission was received	High	2
ARAE-8	Story 1.3: Reject Invalid Uploads	student	to be told immediately when my file is not a valid ZIP	I can fix it and resubmit before the deadline	Medium	3
Epic total						10

Epic 2: Automated Evaluation Pipeline (ARAE-2)

Automatically queue uploaded submissions and run the configured syntax checks and unit test suites, recording the outcomes. (Lab 1: FR-001, FR-003, UC-02, UC-03)

Key	User Story	As a	I want	So that	Priority	Story Points
ARAE-9	Story 2.1: Automatic Evaluation Queue	faculty evaluator	uploaded submissions to be placed in an evaluation queue automatically	no submission is missed or has to be queued by hand	High	5
ARAE-10	Story 2.2: Run Syntax Checks	faculty evaluator	the system to run the configured syntax checks on each submitted project	basic code errors are detected without manual inspection	High	5
ARAE-11	Story 2.3: Run Unit Test Suites	faculty evaluator	the system to execute the pre-configured unit test suite on each submission and record the results	functional correctness is assessed consistently for every student	High	8
ARAE-12	Story 2.4: Evaluation Failure Handling	faculty evaluator	to see a failure status when the tests cannot be executed or the ZIP is un-processable	I can review the failure and take the appropriate action	Medium	3

Epic total	21
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Epic 3: Rubric Scoring (ARAE-3)

Evaluate each submission against the configured rubric and generate an itemized rubric score breakdown. (Lab 1: FR-001, UC-04)

Key	User Story	As a	I want	So that	Priority	Story Points
ARAE-13	Story 3.1: Configure Evaluation Rubric	faculty evaluator	to configure the rubric items and their weights for an assignment	submissions are scored against my grading criteria	High	5
ARAE-14	Story 3.2: Itemized Rubric Score Breakdown	faculty evaluator	the system to calculate a score for each rubric item and generate an itemized breakdown	grading is transparent and consistent	High	8
Epic total						13

Epic 4: Peer Review Assignment (ARAE-4)

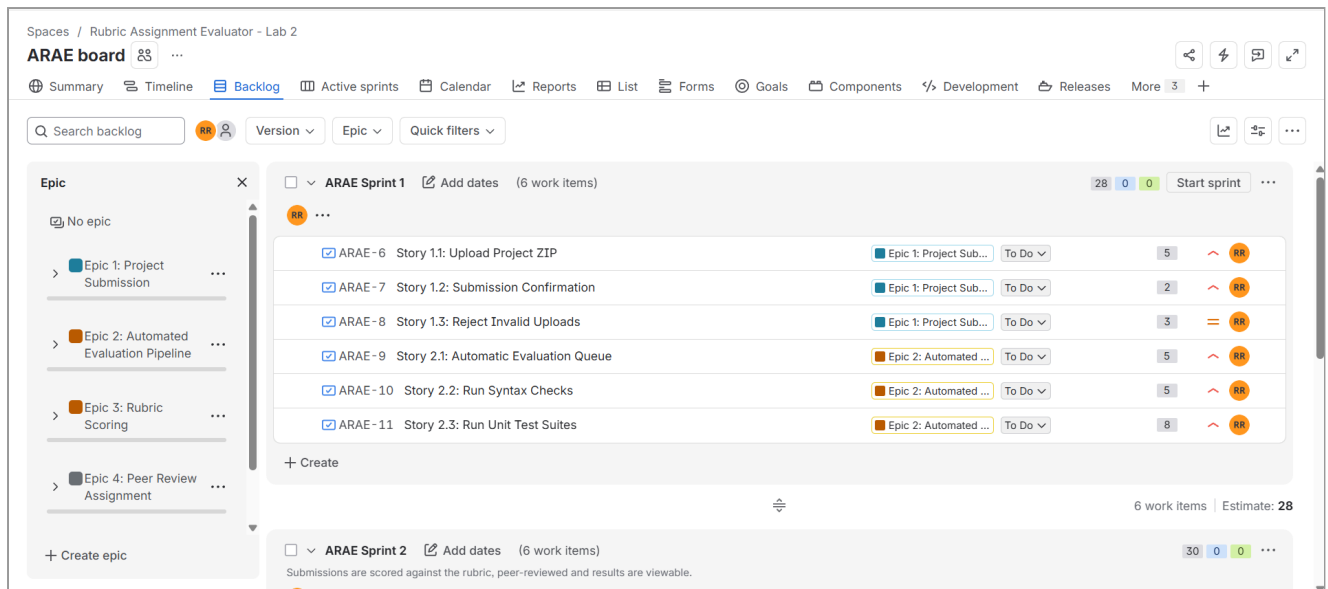
Automatically assign submitted projects to students for peer review without manual distribution by faculty. (Lab 1: FR-004, UC-05)

Key	User Story	As a	I want	So that	Priority	Story Points
ARAE-15	Story 4.1: Automatic Peer Review Assignment	faculty evaluator	submitted projects to be assigned automatically to eligible student reviewers	I do not have to distribute submissions manually	High	8
ARAE-16	Story 4.2: Peer Review Notification	student	to be notified when a project is assigned to me for peer review	I can complete my review on time	Low	2
Epic total						10

Epic 5: Evaluation Results Viewing (ARAE-5)

Allow authorized students and faculty evaluators to view test results and the itemized rubric score breakdown of a completed submission. (Lab 1: FR-005, UC-06)

Key	User Story	As a	I want	So that	Priority	Story Points
ARAE-17	Story 5.1: Student Views Evaluation Results	student	to view the test results and itemized rubric scores for my completed submission	I understand how my project was graded	Medium	3
ARAE-18	Story 5.2: Faculty Views Evaluation Results	faculty evaluator	to view the test results and rubric score breakdown of any evaluated submission	I can review and finalize grades	Medium	3
ARAE-19	Story 5.3: Restrict Results to Authorized Users	student	my evaluation results to be visible only to me and to faculty evaluators	my grades stay private	Medium	5
Epic total						11



Jira backlog with Epics and User Stories

2. Backlog Prioritization and Story Points

Stories are ranked in the backlog in the order below. Priority reflects user impact; story points use the Fibonacci scale (1, 2, 3, 5, 8, 13) for relative effort, complexity and uncertainty.

Rank	User Story	Priority	Story Points	Planned in
1	Story 1.1: Upload Project ZIP	High	5	Sprint 1
2	Story 1.2: Submission Confirmation	High	2	Sprint 1
3	Story 1.3: Reject Invalid Uploads	Medium	3	Sprint 1
4	Story 2.1: Automatic Evaluation Queue	High	5	Sprint 1
5	Story 2.2: Run Syntax Checks	High	5	Sprint 1
6	Story 2.3: Run Unit Test Suites	High	8	Sprint 1
7	Story 2.4: Evaluation Failure Handling	Medium	3	Sprint 2
8	Story 3.1: Configure Evaluation Rubric	High	5	Sprint 2
9	Story 3.2: Itemized Rubric Score Breakdown	High	8	Sprint 2
10	Story 4.1: Automatic Peer Review Assignment	High	8	Sprint 2
11	Story 4.2: Peer Review Notification	Low	2	Backlog
12	Story 5.1: Student Views Evaluation Results	Medium	3	Sprint 2
13	Story 5.2: Faculty Views Evaluation Results	Medium	3	Sprint 2
14	Story 5.3: Restrict Results to Authorized Users	Medium	5	Backlog
Total			65	

Spaces / Rubric Assignment Evaluator - Lab 2

ARAE board

SummaryTimelineBacklogActive sprintsCalendarReportsListFormsGoalsComponentsDevelopmentReleasesMore3

Search backlogVersionEpicQuick filters

Epic

No epic

Epic 1: Project Submission

Epic 2: Automated Evaluation Pipeline

Epic 3: Rubric Scoring

Epic 4: Peer Review Assignment

Create epic

ARAE-12Story 2.4: Evaluation Failure HandlingEpic 2: Automated ...To Do3RR

ARAE-13Story 3.1: Configure Evaluation RubricEpic 3: Rubric Sco...To Do5RR

ARAE-14Story 3.2: Itemized Rubric Score BreakdownEpic 3: Rubric Sco...To Do8RR

ARAE-15Story 4.1: Automatic Peer Review AssignmentEpic 4: Peer Review...To Do8RR

ARAE-17Story 5.1: Student Views Evaluation ResultsEpic 5: Evaluation ...To Do3RR

ARAE-18Story 5.2: Faculty Views Evaluation ResultsEpic 5: Evaluation ...To Do3RR

Create

6 work items | Estimate: 30

Backlog (2 work items)

ARAE-16Story 4.2: Peer Review NotificationEpic 4: Peer Review...To Do2RR

ARAE-19Story 5.3: Restrict Results to Authorized UsersEpic 5: Evaluation ...To Do5RR

Story point assignments

3. Sprints

ARAE Sprint 1

Goal: Students can submit projects and the system queues and tests them automatically.

Duration: 1 week **State:** Completed **Committed:** 28 points **Completed:** 28 points

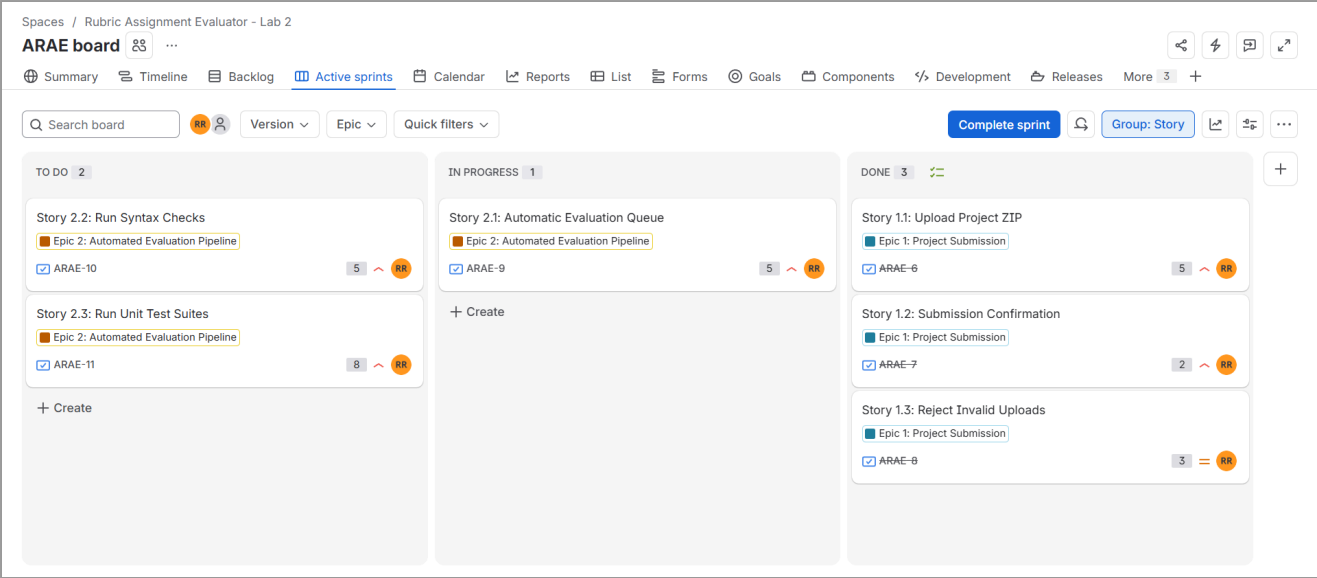
Key	User Story	Story Points	Status
ARAE-6	Story 1.1: Upload Project ZIP	5	Done
ARAE-7	Story 1.2: Submission Confirmation	2	Done
ARAE-8	Story 1.3: Reject Invalid Uploads	3	Done
ARAE-9	Story 2.1: Automatic Evaluation Queue	5	Done
ARAE-10	Story 2.2: Run Syntax Checks	5	Done
ARAE-11	Story 2.3: Run Unit Test Suites	8	Done

ARAE Sprint 2

Goal: Submissions are scored against the rubric, peer-reviewed and results are viewable.

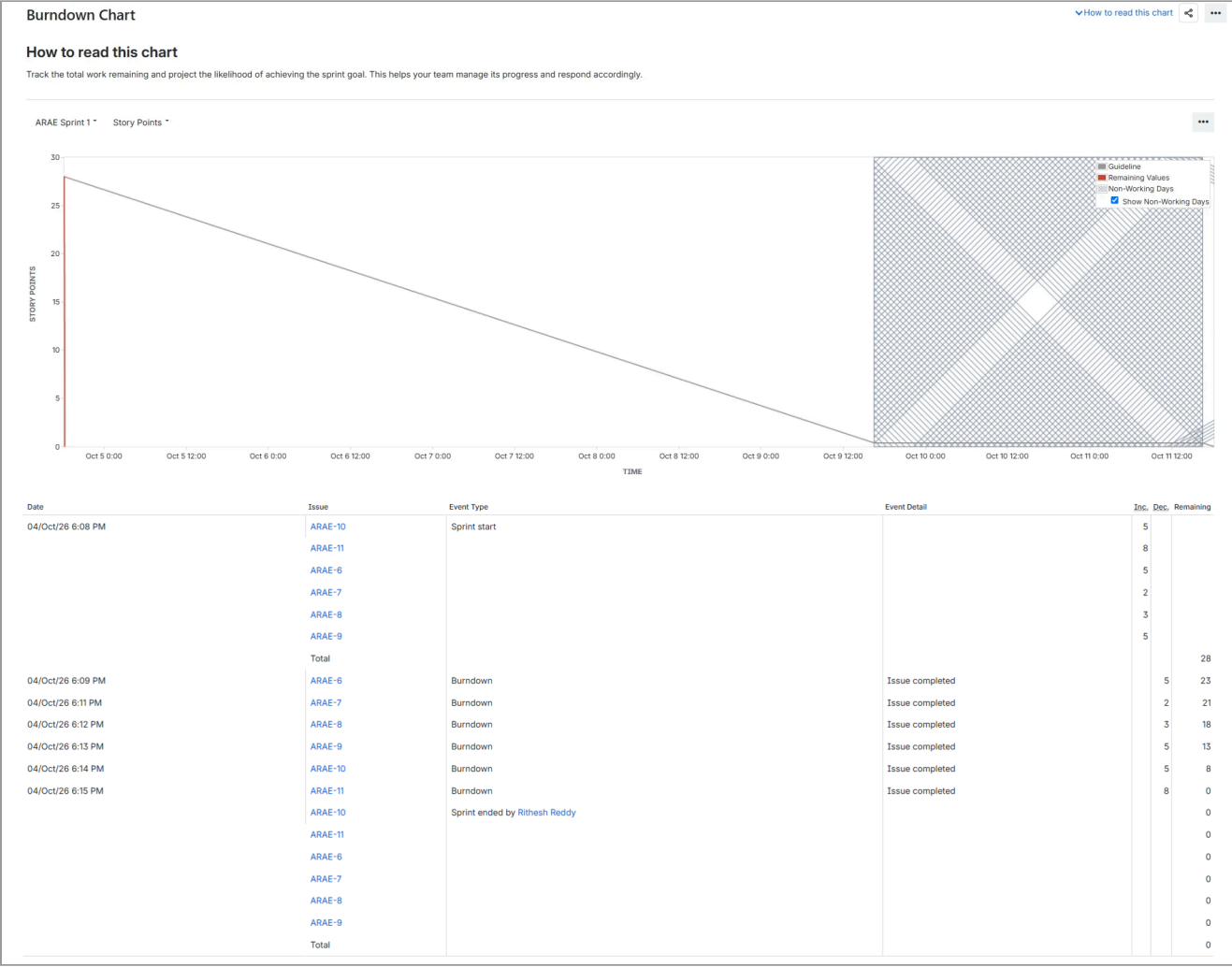
Duration: 1 week **State:** Completed **Committed:** 30 points **Completed:** 30 points

Key	User Story	Story Points	Status
ARAE-12	Story 2.4: Evaluation Failure Handling	3	Done
ARAE-13	Story 3.1: Configure Evaluation Rubric	5	Done
ARAE-14	Story 3.2: Itemized Rubric Score Breakdown	8	Done
ARAE-15	Story 4.1: Automatic Peer Review Assignment	8	Done
ARAE-17	Story 5.1: Student Views Evaluation Results	3	Done
ARAE-18	Story 5.2: Faculty Views Evaluation Results	3	Done

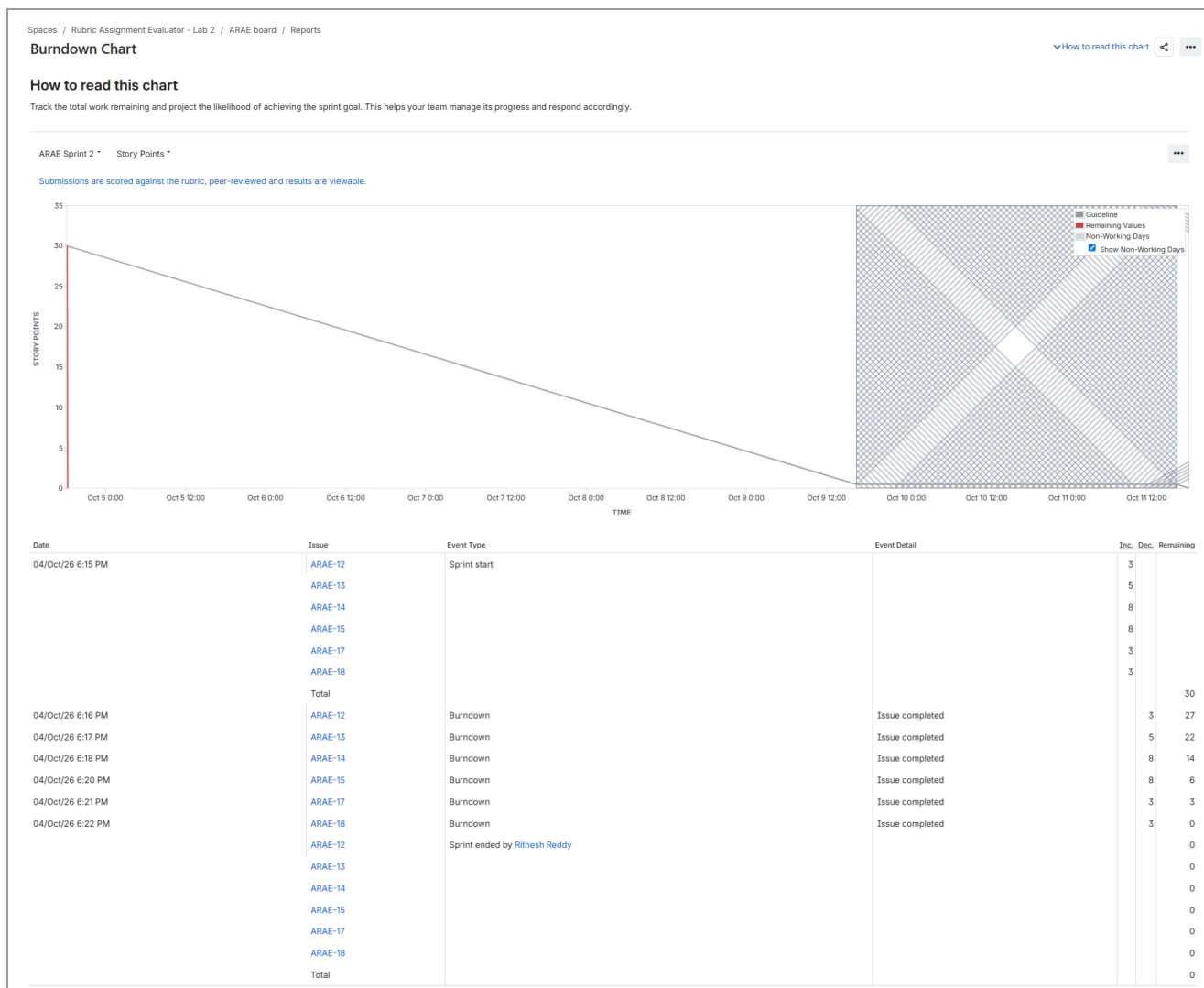


Sprint board (Active Sprint view)

4. Burndown Chart



Burndown Chart - Sprint 1



Burndown Chart - Sprint 2

5. Reflection Questions

1. Did your estimations reflect the actual effort?

Only partly. The story points captured relative effort well: the largest estimates (8 points) went to running the unit test suites, generating the itemized rubric breakdown and automatic peer review assignment, which are the stories with the most logic and uncertainty, while simple stories such as submission confirmation got 2 points. However, because the sprint was simulated, every story was moved from To Do to In Progress to Done at the same pace of about one minute, so an 8-point story took exactly as long as a 2-point story. The estimates therefore could not be validated against real effort. In a real sprint I would compare the time actually spent on each story with its points and adjust future estimates.

2. Was your backlog well-prioritized?

Mostly yes. The backlog was ordered to follow the workflow from Lab 1, so that each story builds on the previous one: upload and confirmation first, then queueing, syntax checks and unit tests, then rubric scoring, peer review and finally viewing results. The High priority stories map to the High priority functional requirements (FR-001 to FR-004) and were placed in the sprints first, while the Low priority peer review notification stayed in the backlog. One weakness is Story 5.3 (restricting results to authorized users): it was left in the backlog as Medium, but since it protects student grades it should probably be rated High and delivered together with the results viewing stories.

3. How did your simulated sprint align with your plan?

The sprints followed the plan exactly in terms of scope. Sprint 1 committed 6 stories worth 28 points and Sprint 2 committed 6 stories worth 30 points; all 12 stories reached Done, nothing was added or removed during the sprints and nothing spilled over. Two stories worth 7 points remain in the backlog for a future sprint. The timing did not match the plan: each sprint was planned for one week but was completed in about seven minutes because the work was simulated. A real sprint would also have blockers, stories stuck in progress and possibly unfinished work carried over, none of which appeared here.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart shows the remaining story points dropping in steps to zero right at the start of each sprint, far below the grey guideline, and scope stays flat with no scope creep. If this were a real team, finishing that far ahead of the guideline would mean the team committed to much less work than it has capacity for and should take on more in sprint planning. The two sprints give a velocity of 28 and 30 points, about 29 on average, which would normally be used to plan the next sprint. Here that number only reflects what was planned, not real capacity, because the stories were moved without real work being done. The main lesson is that the chart is only meaningful when stories are updated as the work actually happens.